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SUBJECT : AI ASSISTED CODING.

ASSIGNMENT : 13.3

Task Description #1 – Remove Repetition

Task: Provide AI with the following redundant code and ask it to refactor

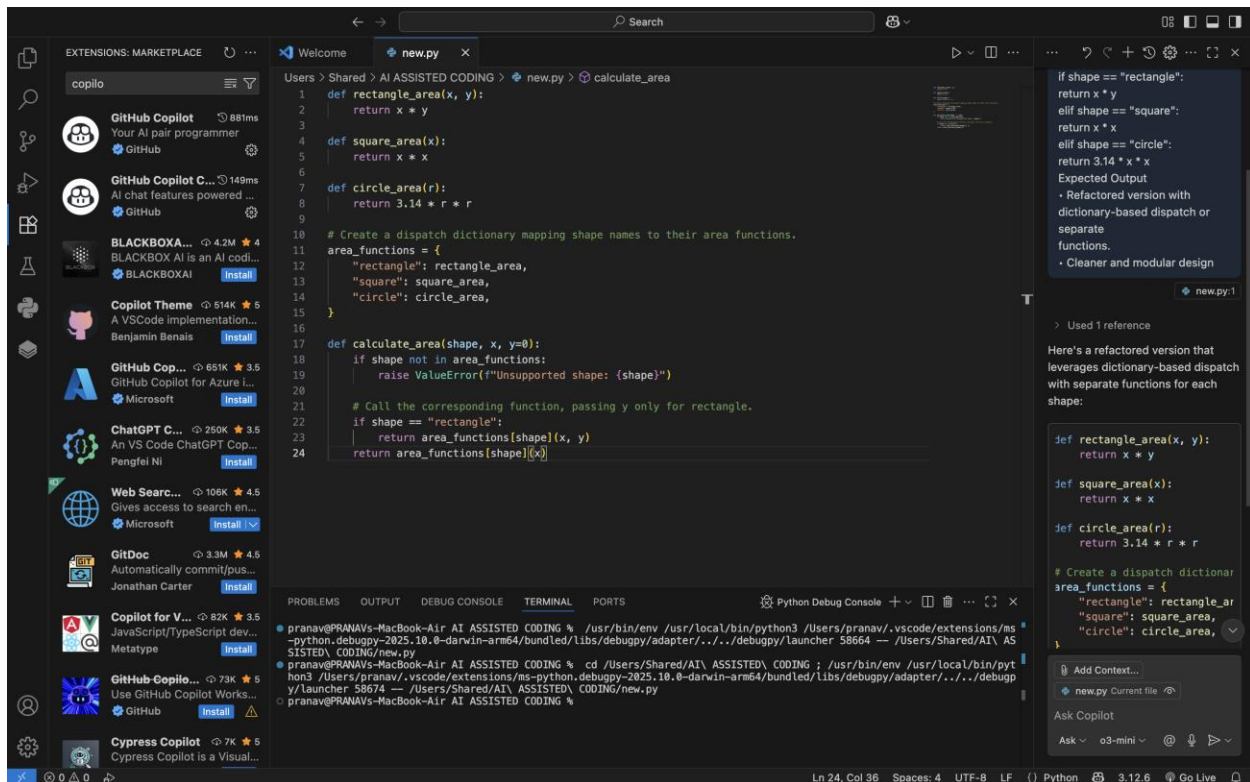
Python Code

```
def calculate_area(shape, x, y=0):  
    if shape == "rectangle":  
        return x * y  
    elif shape == "square":  
        return x * x  
    elif shape == "circle":  
        return 3.14 * x * x
```

Expected Output

- Refactored version with dictionary-based dispatch or separate functions.
- Cleaner and modular design.

CODE GENERATED :



PROMPT :

Write the following redundant code and to refactor the following code

OUTPUT :



Task Description #2 – Error Handling in Legacy Code

Task: Legacy function without proper error handling

Python Code

```
def read_file(filename):
```

```
    f = open(filename, "r")
```

```
    data = f.read()
```

```
    f.close()
```

```
    return data
```

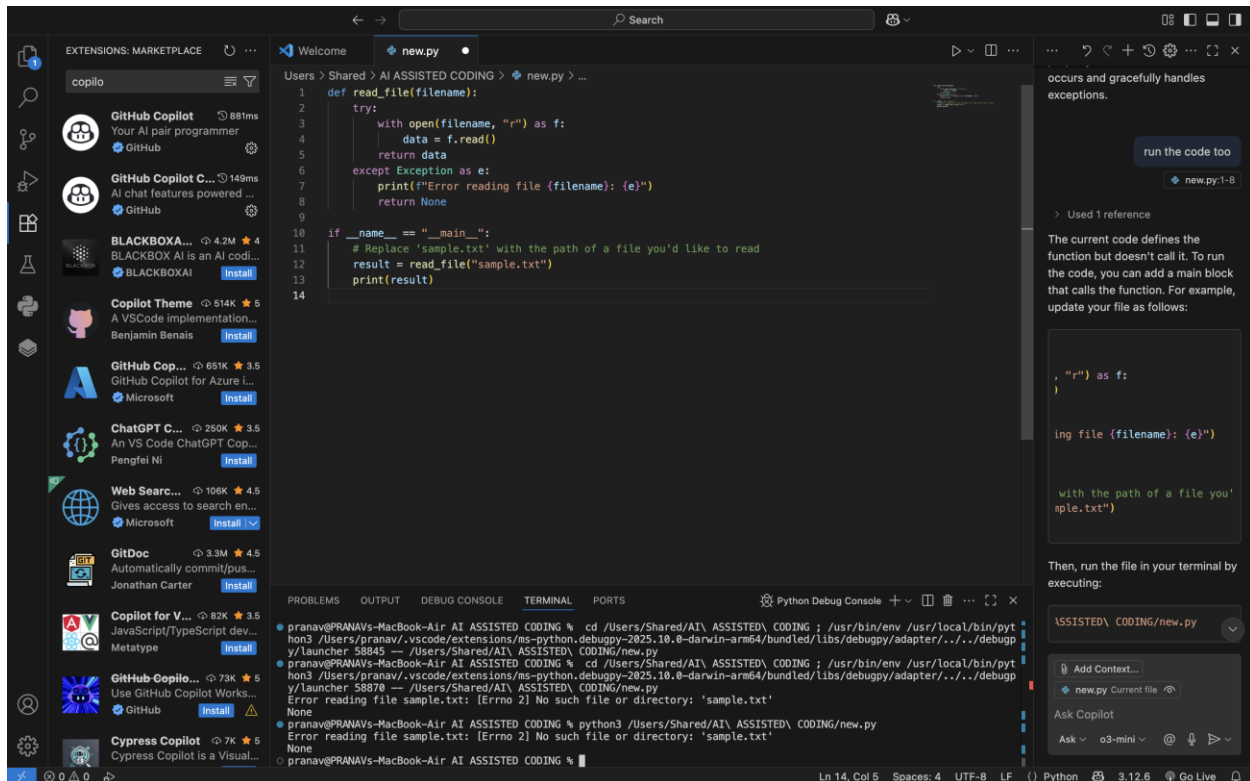
Expected Output:

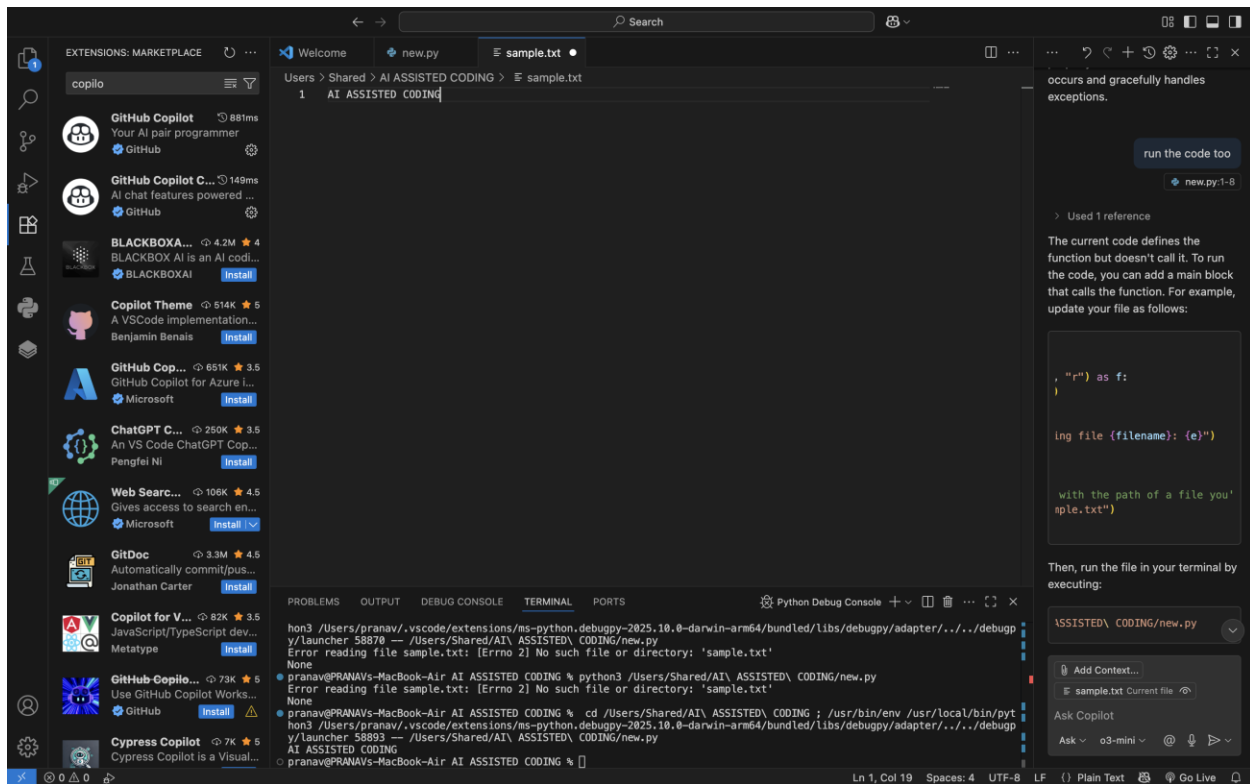
AI refactors with with open() and try-except

PROMPT GIVEN :

To rewrite the legacy function with proper error handling

CODE GENERATED :





OUTPUT GENERATED :



Task Description #3 – Complex Refactoring

Task: Provide this legacy class to AI for readability and modularity improvements:

Python Code

class Student:

def __init__(self, n, a, m1, m2, m3):

self.n = n

self.a = a

self.m1 = m1

self.m2 = m2

```

self.m3 = m3
def details(self):
print("Name:", self.n, "Age:", self.a)
def total(self):
return self.m1+self.m2+self.m3

```

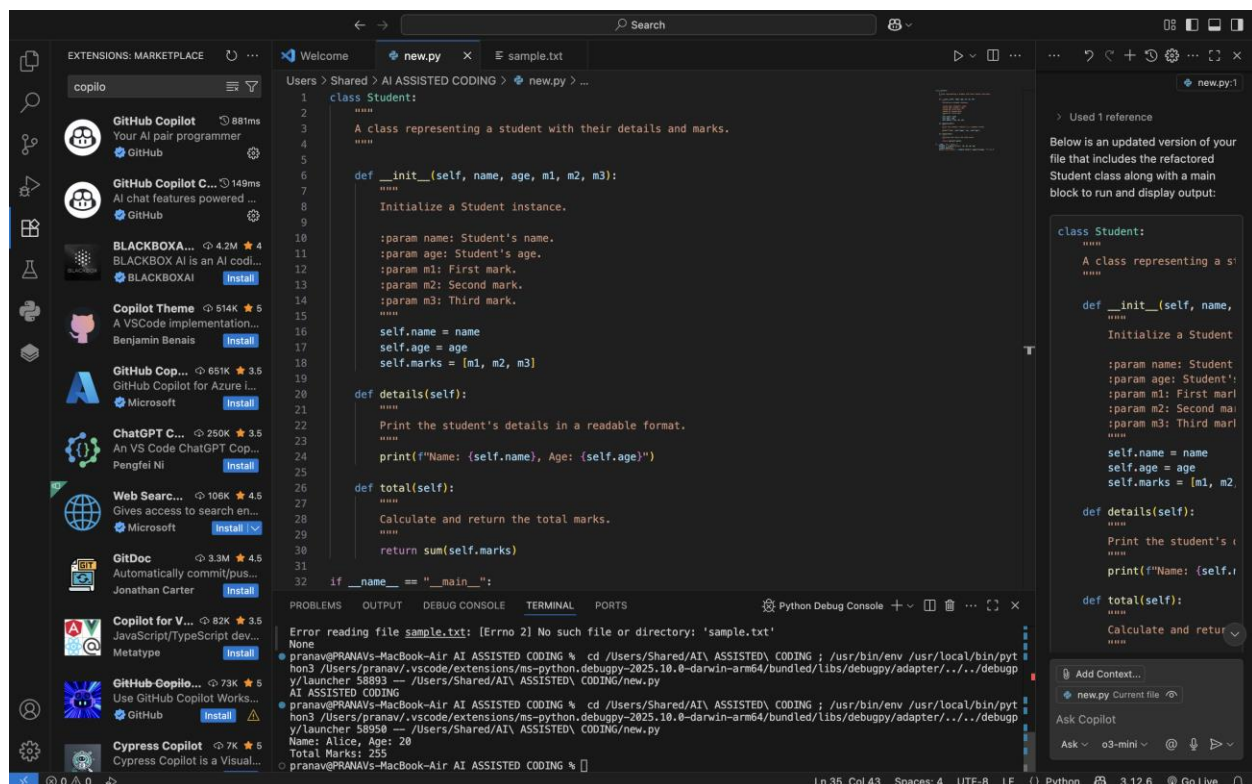
Expected Output:

- AI improves naming (name, age, marks).
- Adds docstrings.
- Improves print readability.
- Possibly uses sum(self.marks) if marks stored in a list

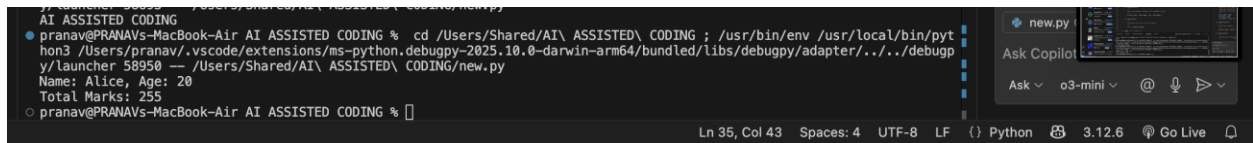
PROMPT USED :

To improve the given code and add docstrings for better readability and possible usage of sum (self.marks) in a list

CODE GENERATED :



OUTPUT GENERATED :

A screenshot of a VS Code terminal window. The terminal shows a prompt 'pranav@PRANAVs-MacBook-Air AI ASSISTED CODING %' followed by a command 'cd /Users/Shared/AI ASSISTED CODING ; /usr/bin/env /usr/local/bin/python3 /Users/pranav/.vscode/extensions/ms-python.debugpy-2025.10.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 58950 -- /Users/Shared/AI ASSISTED CODING/new.py'. The output shows 'Name: Alice, Age: 20' and 'Total Marks: 255'. The terminal status bar at the bottom indicates 'Ln 35, Col 43', 'Spaces: 4', 'UTF-8', 'LF', and the language is 'Python' with version '3.12.6'. A sidebar on the right shows a file explorer with 'new.py' and a chat window with 'Ask Copilot'.

Task Description #4 - Inefficient Loop Refactoring

Task: Refactor this inefficient loop with AI help

Python Code

```
nums = [1,2,3,4,5,6,7,8,9,10]
```

```
squares = []
```

```
for i in nums:
```

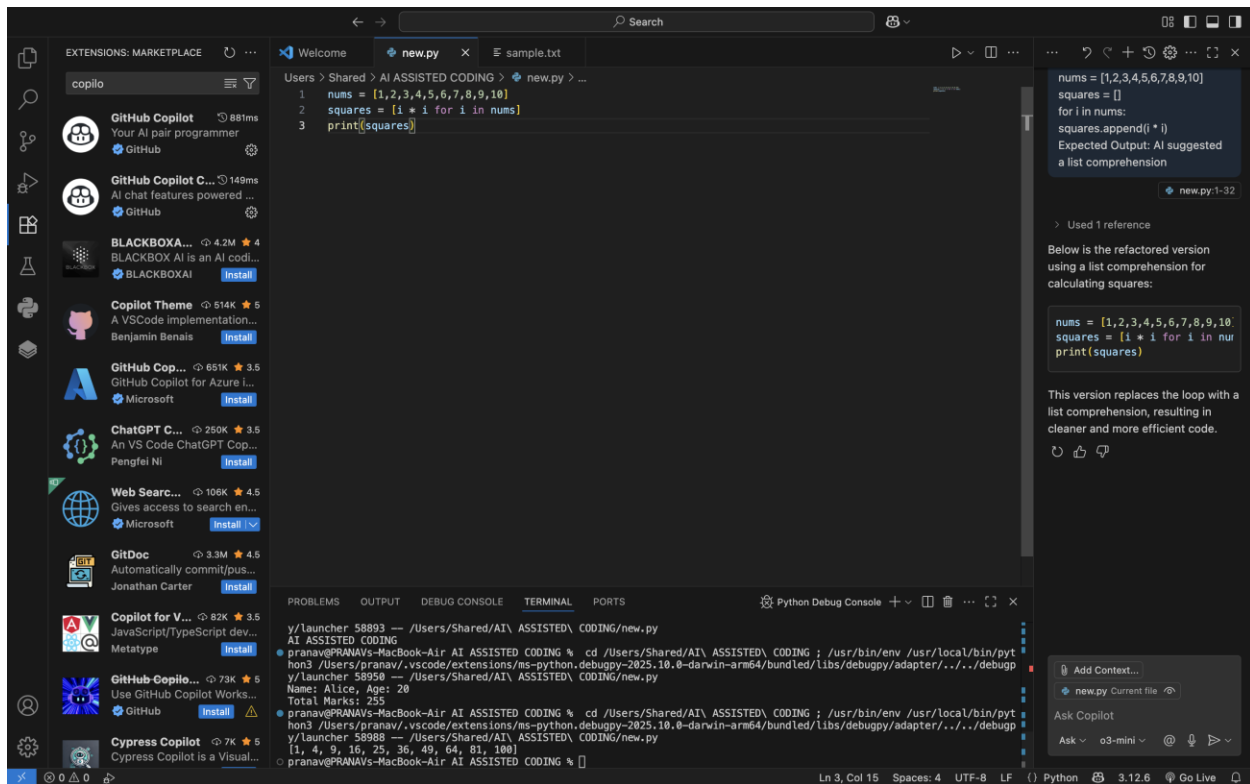
```
    squares.append(i * i)
```

Expected Output: AI suggested a list comprehension

PROMPT USED :

To refactor the given loop and list comprehension

CODE GENERATED :



OUTPUT:

